

Degree vs. Skill: Are FJWU Students Truly Job-Ready?

A Data Analytics Perspective on Skills vs Academic Performance

Capstone Project Report

Department of Computer Science
Fatima Jinnah Women University, Rawalpindi
Batch 2025

Project Team

Name	Roll Number	Role	Responsibilities
Insa Saleem	2025-BCS-043	Data Engineering	Data collection, cleaning, integration, overall analysis
Eman Fatima	2025-BCS-025	Statistical Analysis	Exploratory Data Analysis, descriptive statistics, correlation analysis
Zunaira Tariq	2025-BCS-095	NLP & Applied Analytics	Sentiment analysis, keyword extraction, behavioral analysis
Maria Yasmeen	2025-BCS-059	Project Lead	Documentation, report writing, research coordination
Isra Asif	2025-BCS-045	Interface & Visualization	Dashboard design, UI development, result consolidation

Table of Contents

1. Introduction
2. Problem Statement
3. Objectives
4. Scope of the Project
5. Literature Review
6. Methodology
7. Data Collection
8. Data Cleaning and Preprocessing
9. Exploratory Data Analysis
10. Statistical Analysis
11. NLP and Sentiment Analysis
12. Dashboard and Interface Design
13. Key Findings
14. Challenges Faced
15. Conclusion
16. Future Recommendations
17. Tools and Technologies Used
18. References

Abstract

This capstone project investigates whether Computer Science students at Fatima Jinnah Women University (FJWU) are genuinely job-ready beyond academic achievement. The research focuses on the relationship between CGPA, technical skills, industry exposure, certifications, and practical readiness for employment.

A survey dataset containing responses from 108 students was collected and analyzed using data analytics, statistical methods, and Natural Language Processing (NLP) techniques. The project aimed to identify gaps between academic performance and industry expectations.

The findings reveal that high academic scores alone are not sufficient indicators of professional readiness. Students with internships, certifications, GitHub portfolios, and project-based learning experiences demonstrated significantly higher confidence and practical competence compared to students relying only on coursework.

The project also highlights major deficiencies in communication skills, DevOps knowledge, project management, and cloud technologies — areas that are increasingly demanded in the software industry.

The final outcome includes statistical analysis, data visualizations, NLP-based sentiment insights, and an interactive dashboard interface to present findings in an accessible and modern format.

1. Introduction

The technology industry is evolving rapidly, demanding graduates who possess not only academic knowledge but also practical skills, adaptability, communication abilities, and real-world exposure. In many universities, students focus heavily on CGPA and examinations while industry recruiters increasingly prioritize practical experience and problem-solving capabilities.

This project explores the gap between academic performance and practical employability among FJWU Computer Science students. Through survey-based analytics, the project evaluates whether students feel prepared for employment and identifies the key factors influencing job readiness.

The research integrates statistical analysis, visualization techniques, and Natural Language Processing to transform student feedback into measurable insights.

2. Problem Statement

Many Computer Science students achieve strong academic results but still struggle to secure internships and employment opportunities due to insufficient practical exposure.

The core problem addressed in this project is:

“Does academic performance alone determine job readiness, or do practical skills and industry exposure play a more significant role?”

The project investigates:

- The relationship between CGPA and practical skills
- Student confidence regarding employment
- Industry exposure through internships and portfolios
- Skills demanded by industry but missing in university coursework
- Sentiments and concerns expressed by students regarding career readiness

3. Objectives

The main objectives of this project are:

1. To analyze the relationship between CGPA and practical skills.
2. To measure students' self-rated job readiness.
3. To identify skill gaps between academia and industry.
4. To analyze internship participation and professional exposure.
5. To apply NLP techniques for sentiment analysis on student responses.
6. To develop an interactive dashboard for presenting analytical insights.
7. To provide recommendations for improving student employability.

4. Scope of the Project

The scope of this project includes:

- Survey-based analysis of FJWU Computer Science students
- Statistical and visual analysis of academic and skill-related data
- NLP processing of open-ended textual responses
- Dashboard development for interactive result presentation
- Identification of industry-relevant skill gaps

The project does not include:

- External university datasets
- Recruitment company datasets
- Longitudinal employment tracking
- AI-based predictive hiring systems

5. Literature Review

Modern recruitment practices emphasize practical skills alongside academic qualifications. Studies in higher education analytics suggest that students with internships, collaborative projects, certifications, and technical portfolios demonstrate stronger employability outcomes.

Research also highlights the increasing importance of:

- Cloud computing
- Agile methodologies
- Communication skills
- DevOps practices
- Real-world software engineering workflows

Several studies reveal that universities often focus more on theoretical instruction than practical industry preparation. This mismatch creates a “skills gap” between graduates and employer expectations.

The current project contributes to this discussion by specifically analyzing the condition of Computer Science students at FJWU.

6. Methodology

The project followed a structured data analytics workflow.

Step 1: Survey Design

A Google Form survey was designed to collect responses regarding:

- CGPA
- Technical skills
- Certifications
- Internship experience
- Career confidence
- Project experience
- Career concerns

A total of 108 student responses were collected.

Step 2: Data Collection

The survey data was exported into spreadsheets for analysis and preprocessing.

Step 3: Data Cleaning

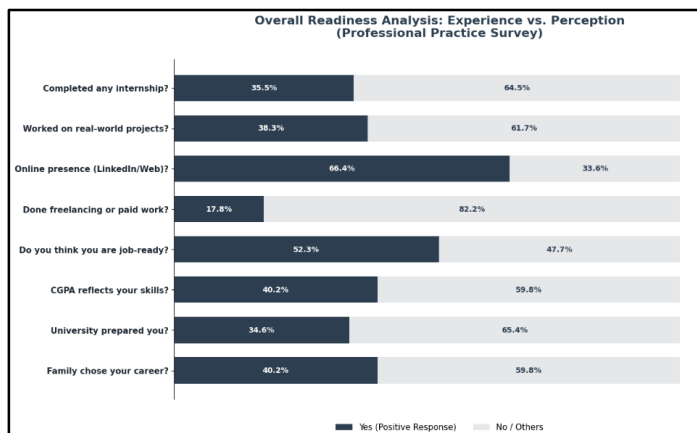
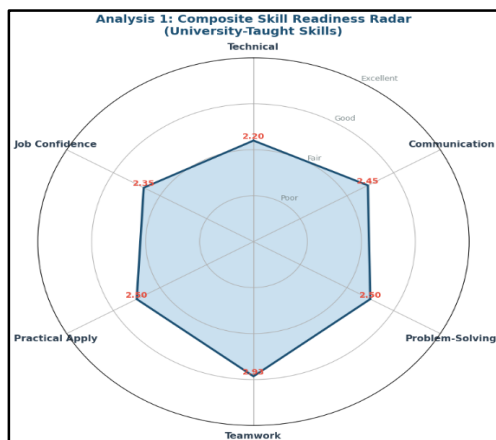
The dataset underwent:

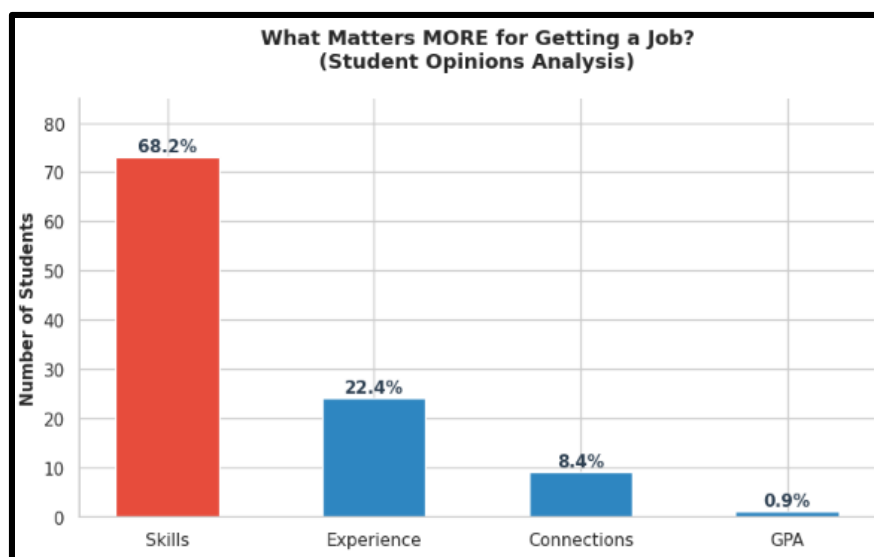
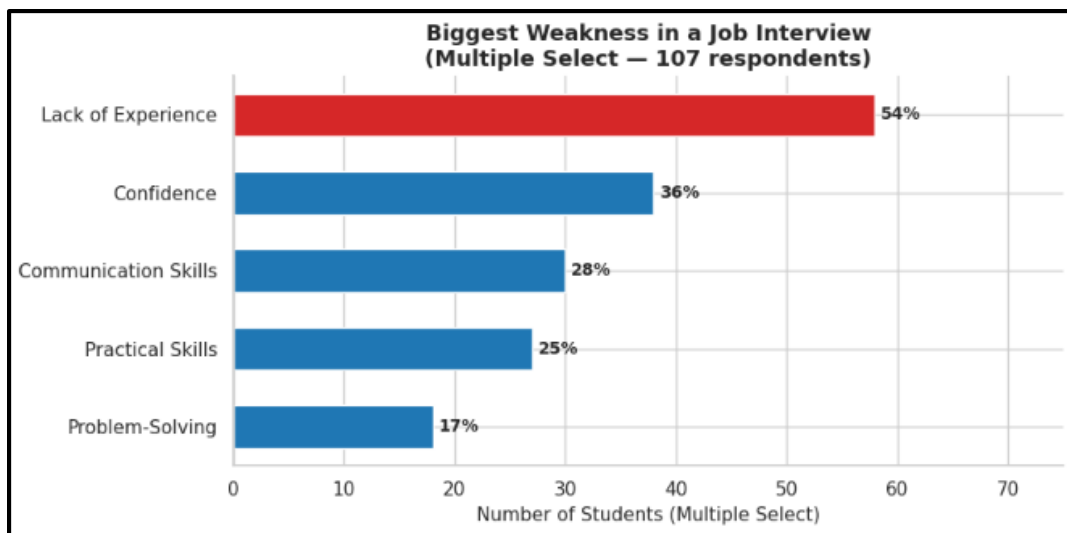
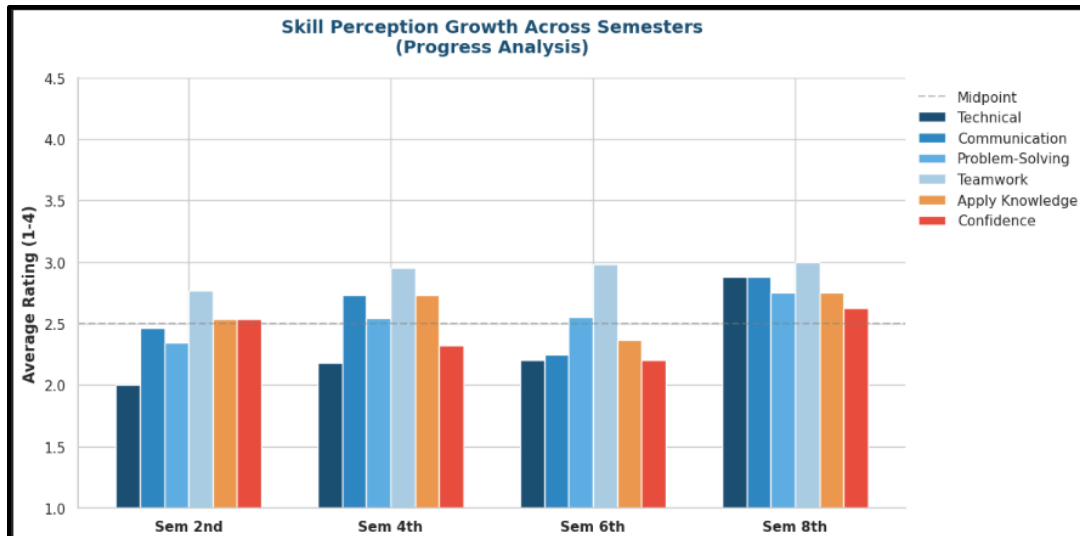
- Missing value handling
- Duplicate removal
- Text normalization
- Formatting standardization
- Structural cleaning

Step 4: Exploratory Data Analysis

EDA techniques were applied to:

- Understand distributions
- Identify trends
- Detect relationships
- Analyze readiness patterns

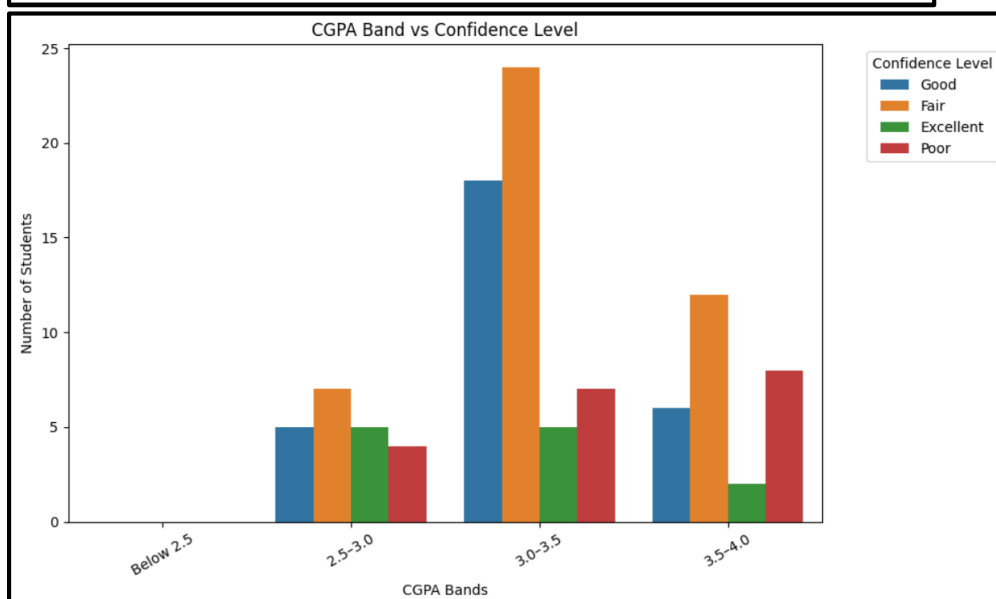
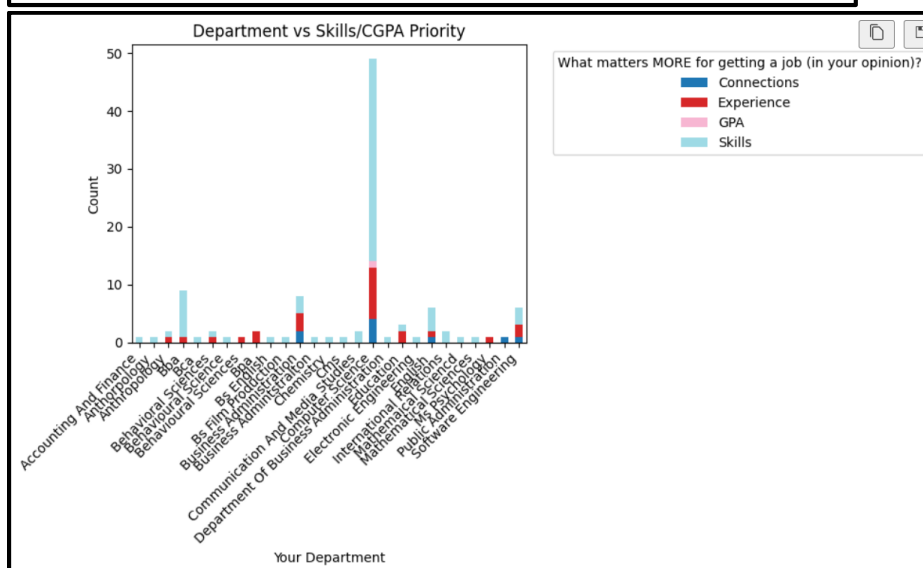
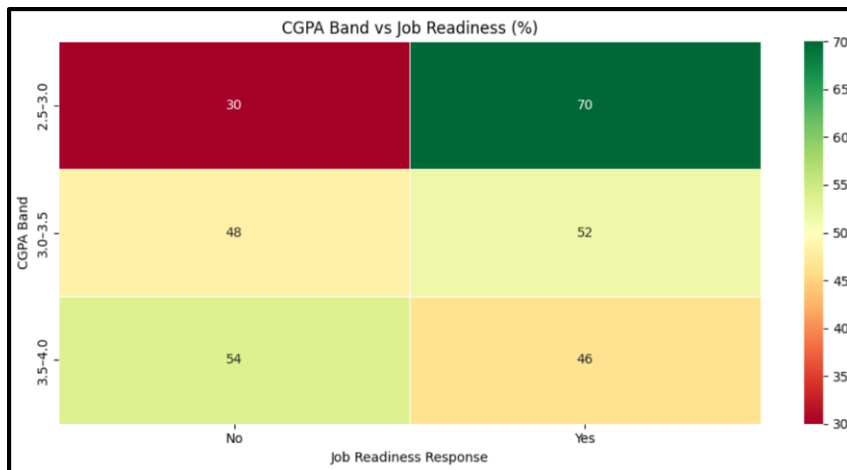


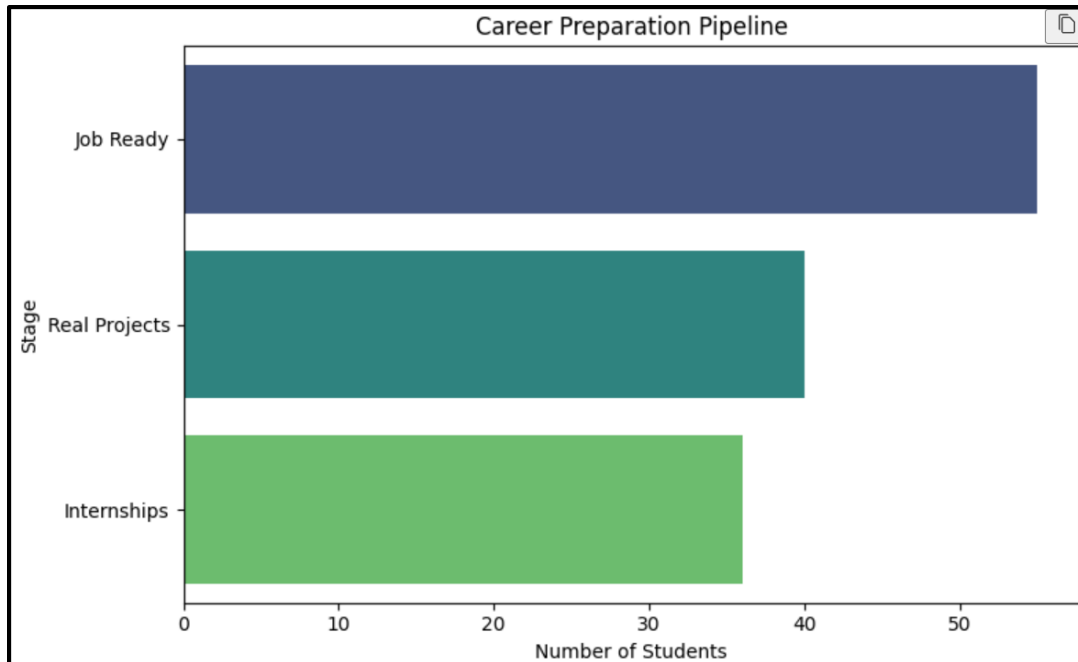


Step 5: Statistical Analysis

Statistical analysis included:

- Correlation analysis
- Distribution analysis
- Comparative skill assessment
- Internship impact evaluation





Step 6: NLP Analysis

Open-ended responses were processed using NLP methods including:

- Sentiment analysis
- Keyword extraction
- Behavioral analysis

Step 7: Dashboard Development

An HTML-based dashboard was developed using:

- Chart.js
- Interactive visualizations
- Responsive UI components
- Modern analytics design principles

7. Data Collection

The dataset consisted of 108 responses collected from Computer Science students at Fatima Jinnah Women University.

Major Variables Collected

Category	Examples
Academic Metrics	CGPA, semester

Category	Examples
Technical Skills	Programming, DSA, databases, AI/ML
Professional Exposure	Internships, freelancing, GitHub
Certifications	Online certifications, technical courses
Soft Skills	Communication, teamwork
Career Confidence	Job readiness rating
Open-Ended Responses	Student concerns and feedback

8. Data Cleaning and Preprocessing

The raw survey data contained inconsistencies and formatting variations that required preprocessing.

Cleaning Procedures

- Removal of incomplete records
- Standardization of categorical responses
- Lowercase conversion
- Noise removal from textual responses
- Missing value handling
- Dataset restructuring

Files Used

The project utilized multiple datasets and scripts including:

- FJWU_Data_Fixed_Final.xlsx
- cleaned_data.xlsx
- structurally_cleaned_data.xlsx
- behavioral_integrity_analysis.xlsx
- columns_combined.py
- noise_removal.py
- del_missing_rows.py

9. Exploratory Data Analysis

EDA was performed to understand the overall structure and trends of the dataset.

CGPA Distribution

The majority of students fell within the 3.0–3.5 CGPA range, followed by students in the 3.5–4.0 range.

CGPA Range	Approximate Students
2.0–2.5	8
2.5–3.0	22
3.0–3.5	45
3.5–4.0	33

Self-Rated Job Readiness

Students were asked how prepared they felt for employment.

Readiness Level	Responses
Not Ready	24
Somewhat Ready	38
Moderately Ready	31
Very Ready	15

The data suggests that many students lack confidence regarding their employability.

Technical Skills Analysis

Students rated their proficiency in:

- Programming
- Data Structures and Algorithms
- Databases
- Web Development
- AI/ML
- Communication
- DevOps
- Project Management

Programming skills were relatively strong, while DevOps and project management showed weaker ratings.

Industry Exposure

The analysis showed:

- 68% of students had no internship experience
- Many students lacked portfolios or GitHub activity

- Freelancing participation remained low
- Open-source contribution rates were minimal

10. Statistical Analysis

Statistical analysis was conducted to measure relationships between variables.

Key Statistical Findings

1. CGPA Is Not a Strong Predictor of Job Readiness

Students with higher CGPAs did not always demonstrate stronger practical skills.

2. Internship Experience Improves Confidence

Students with internships rated themselves significantly more job-ready than students without practical exposure.

3. Certifications Positively Impact Skills

Students with external certifications generally demonstrated:

- Better technical awareness
- Higher confidence
- Greater exposure to modern technologies

4. Industry Demand vs Student Readiness Gap

The largest gaps existed in:

- Cloud computing
- Agile methodology
- DevOps
- Communication skills
- Project management

11. NLP and Sentiment Analysis

Natural Language Processing techniques were applied to analyze open-ended survey responses.

NLP Objectives

- Measure student sentiment
- Identify recurring concerns
- Extract keywords from responses

- Detect confidence patterns

Common Student Concerns

Students frequently mentioned:

- Lack of practical projects
- Fear of interviews
- Insufficient real-world exposure
- Lack of confidence
- Limited internship opportunities

Sentiment Results

The majority of responses reflected:

- Neutral sentiment
- Mildly negative sentiment regarding career readiness
- Anxiety about industry competition

Positive responses were mostly associated with:

- Internship experience
- Certifications
- Self-learning
- Personal projects

12. Dashboard and Interface Design

An interactive analytics dashboard was developed to present project findings visually.

Features of the Dashboard

- Responsive design
- Interactive charts
- Modern UI aesthetics
- Real-time visualization rendering
- Professional analytics interface

Technologies Used

- HTML5
- CSS3
- JavaScript
- Chart.js

Visualizations Included

- CGPA distribution chart
- Readiness doughnut chart
- Skills radar chart
- Industry exposure chart
- Gap analysis bar chart

The dashboard was designed to make insights understandable for both technical and non-technical audiences.

13. Key Findings

Major Insights

Academic Performance Alone Is Insufficient

A high CGPA does not guarantee industry readiness.

Internships Significantly Improve Readiness

Students with internships demonstrated stronger confidence and practical awareness.

Certifications Fill Curriculum Gaps

External learning platforms help students gain modern industry-relevant knowledge.

Soft Skills Remain Underdeveloped

Communication and teamwork require greater emphasis in academic programs.

Industry Expectations Are Changing

Employers increasingly demand:

- Practical development experience
- GitHub portfolios
- Collaboration skills
- Cloud and DevOps knowledge
- Agile development understanding

14. Challenges Faced

Several challenges were encountered during project development.

Technical Challenges

- Cleaning inconsistent survey responses
- Standardizing subjective skill ratings
- Managing missing values
- Processing textual feedback for NLP analysis

Research Challenges

- Limited sample size
- Subjective self-assessment bias
- Time constraints for deeper industry validation

Development Challenges

- Designing professional visualizations
- Maintaining dashboard responsiveness
- Integrating multiple analytical modules

15. Conclusion

The project successfully demonstrates that academic performance alone cannot fully determine a student's job readiness.

The analysis shows that practical exposure, certifications, internships, technical projects, and continuous self-learning play a much more significant role in employability.

Many FJWU students possess strong theoretical foundations but require greater exposure to real-world development practices and industry-standard technologies.

The project highlights the urgent need for stronger integration between university curriculum and industry expectations.

Through data analytics, statistical analysis, NLP processing, and dashboard visualization, the project provides meaningful insights into student readiness and future educational improvements.

16. Future Recommendations

For Students

- Build GitHub portfolios
- Participate in internships
- Learn cloud and DevOps tools
- Work on real-world projects

- Improve communication skills

For Universities

- Increase project-based learning
- Introduce industry collaboration programs
- Add practical workshops and bootcamps
- Encourage open-source contribution
- Integrate modern technologies into curriculum

For Future Research

- Expand the survey to multiple universities
- Apply predictive analytics models
- Use machine learning for employability prediction
- Conduct longitudinal career tracking

17. Tools and Technologies Used

Category	Technologies
Data Analysis	Python, Pandas, NumPy
Visualization	Chart.js, Matplotlib
NLP	NLTK, Text Processing Scripts
Dashboard	HTML, CSS, JavaScript
Data Storage	Excel Spreadsheets
Documentation	Jupyter Notebook, Research Writing

Appendix

Project Files Included

Statistical Analysis

- Capstone_Project.ipynb

Data Engineering

- FJWU_Data_Fixed_Final.xlsx

- PP_Overall_Analysis.ipynb

NLP and Behavioral Analysis

- Automated_Student_Analysis_Final.xlsx
- behavioral_integrity_analysis.xlsx
- cleaned_data.xlsx
- structurally_cleaned_data.xlsx
- insights.py

Dashboard Interface

- fjwu_job_readiness.html

Final Remarks

This capstone project reflects the collaborative effort of five Computer Science students working across data engineering, statistical analysis, NLP, interface design, and research documentation.

The project demonstrates how data analytics can be used to evaluate educational outcomes and identify gaps between academic systems and industry expectations.

The findings can help students, educators, and institutions better understand the evolving definition of “job readiness” in the modern technology landscape.